Regression (linear)

$y \in \mathbb{R}$

$y_i = f(x_i)$

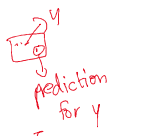
$y_i = w^T x_i$

$\sum_{i=1}^{n} (y_i - w^T x_i)^2 \Rightarrow$ error

$2xx^T w - 2xy$ $\quad x \Rightarrow [\,|\,|\,|\,|\,|\,]_{d \times n}$

$w = (xx^T)^+ xy$

$y \rightarrow$ project y onto span of feature vectors $(f_1 \dots f_d)$



prediction for y

$y_{pred} = X^T w$ ↳ optimal vector

$\rightarrow w = (xx^T)^+ xy$

↳ costly

↳ gradient descent

$2xx^T w - 2xy$

$w \Rightarrow$ intialize

$$\boxed{\begin{array}{l} w^0 \\ w^1 = w^0 - \eta \nabla f(w^0) \end{array}}$$

↳ positive

until convergence $(w^{t+1} != w^t)$

$w^{t+1} = w^t - \eta (2xx^T(w^t) - 2xy)$

↓ learning rate

when will ∇ be 0?

at critical point

if $f \Rightarrow$ convex function, then converges to global min,

generally local minimum

$A \times B = c$ $\quad c_{i,j} = A(i,:) \cdot B(:,j)$

$m \times n$ $\quad n \times p$

$\nabla \Rightarrow 2xx^T w - 2xy$

↳ $d \times d$

↳ $(d \times n) \cdot (n \times d)$

↳ no. of data points

Stochastic gradient Descent

$w \Rightarrow$ initialize $\quad k << n$

for (τ) rounds :

S = sample $(k) \Rightarrow$ datapoints $\quad [\,|\,|\,|\,|\,]$

$\nabla \Rightarrow 2SS^T w - 2Sy$

$(d \times k)(k, d)$

w^0

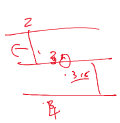
w^1

$w^2 \leftarrow$

$w^3 = w^2 - \eta \nabla f(w^2)$

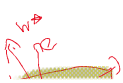
$w^{t+1} = w^t - \eta \nabla (w^t)$

after T rounds $w^T = \frac{1}{t} \sum_{i=1}^{t} w^i$



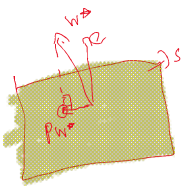
$x_1 \dots x_n \in \mathbb{R}^d$

$w^* \in \mathbb{R}^d$ $\quad e + \rho w^* = w^*$



span (data points)

$w^* \in \mathbb{R}^d$



$\rightarrow$ span (data points)

$x_i \rightarrow$ predict (y_i)

$w^* \qquad P_{w^*}$

$$(x_i^T w^*) = (x_i^T P_{w^*})$$

$$(x_i^T (P_{w^*} + e)) \rightarrow (x_i^T P_{w^*} + \underbrace{x_i^T e}_{0})$$

$w^* \longleftrightarrow P_{w^*}$

pred = pred

$$\sum_{i=1}^{n} (y_i - w^{*T} x_i)^2 = \sum_{i=1}^{n} (y_i - P_{w^*}^T x_i)^2$$

$w^* \Rightarrow$ optimal solution

$P_{w^*} \Rightarrow$ optimal solution

w^*

$w^* = P_{w^*}$

the optimal solution will be in the span (data points)

$$w^* = \begin{bmatrix} | & | & | & | \end{bmatrix} \begin{bmatrix} a \end{bmatrix}$$

$$w^* = \sum_{i=1}^{n} a_i x_i = X a$$

$$\boxed{w^* = Xa} \quad \text{for some } a$$

$$\phi((x, y)) \Rightarrow (x^2, y^2, x, y, xy, 1)$$

y

linear regression

Kernel Regression

hypothesis $\Rightarrow$ relationship $(X, Y) \Rightarrow$ non-linear

$\phi : \mathbb{R}^d \rightarrow \mathbb{R}^D \quad D > d$

$$\phi X = [\phi x_1, \phi x_2, \ldots, \phi x_n]$$

$x_1 \rightarrow \phi x_1$
$x_2 \rightarrow \phi x_2$
$x_3 \rightarrow \vdots$
$x_4 \rightarrow \vdots$
$\vdots$
$x_n \rightarrow \phi x_n$

Y & ϕX

do linear regression

$$w = (\phi X \cdot \phi X^T)^+ \phi X Y$$

shortcut $\Rightarrow$ kernel

$w \Rightarrow$ solution

$$w = (\phi X \phi X^T)^{\pm} \phi X Y$$

$$w = \phi X \alpha$$

I don't know

$$\phi X \alpha = (\phi X \phi X^T)^{\dagger} \phi X y$$

$$(\phi X \phi X^T)\, \phi X \alpha = (\phi X \phi X^T)\, (\phi X \phi X^T)^{\dagger}\, \phi X y$$

$$(\phi X)(\phi X^T \phi X)\alpha = (\phi X) y$$

$$\underbrace{(\phi X^T)(\phi X)}_{K}\, \underbrace{(\phi X^T \phi X)}_{K}\, \alpha = \underbrace{\phi X^T \phi X}_{K}\, y$$

$\phi X^T \phi X \Rightarrow$ kernel matrix

$$\begin{bmatrix} - \phi x_1 - \\ - \phi x_2 - \\ \vdots \\ - \phi x_n - \end{bmatrix} \begin{bmatrix} | & | & & | \\ \phi x_1 & \phi x_2 & \cdots & \phi x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} k(x_1, x_1) \\ k(x_1, x_2) \end{bmatrix}$$

$$\phi x_1^T \phi x_2 = k(x_1, x_2)$$

$$K^2 \alpha = K y$$

$$K\alpha = y$$

$$\boxed{\alpha = K^{\dagger} y}$$

$$\underset{n \times n}{[\ \ ]}\ \underset{n \times 1}{(\ \ )}$$

$n \times 1$

$$w = \underset{d \times n}{X}\ \underset{n \times 1}{\alpha}$$

$K \Rightarrow n \times n$

$k_{ij} = k(x_i, x_j)$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$\phi X \alpha$

need not compute w

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}^{\dagger} \Big($$

$x_t \to y_t$

$$x_t \to \phi x_t$$

max

$$w^T \phi x_t$$

$$w = \phi X \alpha \qquad (\alpha_1\ \alpha_2\ \ldots\ \alpha_n)$$

$$w^T = \alpha^T \phi X^T$$

$$= \Big(\sum^{n} \ldots \phi x_i^T\Big)$$

ϕX_1^T

ϕX_2^T —

$\vdots$

ϕX_{10}^T —

$(a+b)c = ac+bc$

$$(w)^T \phi x_t$$

$$\left[\sum_{i=1}^{n} \alpha_i \phi x_i^T\right] \phi x_t$$

$$= \left(\sum_{i=1}^{n} \alpha_i \phi x_i^T\right)$$

$n \times n$

$$= \left[\alpha_1 c\right.$$

$$\sum_{i=1}^{n} \alpha_i \phi x_i^T \phi x_t$$

$$\boxed{\sum_{i=1}^{n} \underbrace{\alpha_i}_{\text{scalar}} \underbrace{k(x_i, x_t)}_{\text{similarity b/w } x_i \,\&\, x_t}}$$

$$y_{t(\text{pred})} = \alpha_1 k(x_1, x_t) + \alpha_2 k(x_2, x_t) + \cdots + \alpha_n k(x_n, x_t)$$

weight

$$\begin{bmatrix} \phi x_1^T \\ \vdots \\ \phi x_n^T \end{bmatrix}$$

$$(a+b)c = ac+bc$$

$$\phi x_1^T + \alpha_2 \phi x_2^T + \cdots + \alpha_n \phi x_n^T \Big] \phi x_t$$